PART II: PATTERNS

Topic 17: Stationarity — The Stable Baseline

The Core Concept

A time series is **stationary** if its key statistical properties — mainly its average (mean) and variability — remain constant over time. It has no trend and no growing or shrinking swings.

- **Stationary:** The data bounces within a consistent band around a fixed average. It looks the same wherever you look.
- **Non-stationary:** The average is moving (trend) or the swings are changing in size (growing volatility or seasonality).

Why It Matters: You Cannot Predict a Moving Target

Most advanced forecasting tools require stationary data. A non-stationary series must first be "stabilized" before predictions can be trusted.

Think of a basketball analogy:

- **Stationary player:** Scores around 20 points every game. Easy to predict: expect ~20 next game.
- **Non-stationary player:** Improving every week (trend) or scores 40 in winter and 5 in summer (seasonality). A simple average is useless as a prediction.

The Example

Note: All data in this tutorial series are synthetic, created for teaching purposes only. They do not represent real-world incidents or any specific jurisdiction.

Year	Period	Precinct North (UoF)	Precinct South (UoF)	North Arrests
2021	Jan–Mar	28, 26, 30	25, 24, 26	~500 avg
	Apr–Jun	32, 35, 40	24, 25, 23	
	Jul–Sep	42, 38, 34	26, 25, 24	
	Oct–Dec	30, 28, 25	25, 26, 24	
2022	Jan–Mar	22, 20, 24	24, 26, 25	~450 avg
	Apr–Jun	26, 28, 32	23, 25, 24	
	Jul–Sep	35, 30, 28	25, 26, 24	
	Oct–Dec	24, 22, 18	24, 25, 23	
2023	Jan–Mar	12, 10, 14	25, 22, 28	~410 avg
	Apr–Jun	15, 18, 22	24, 26, 24	
	Jul	25	27	700
	Aug	**(Missing)**	23	680
	Sep	19	22	500
	Oct	16	**85 ★ Outlier**	420
	Nov–Dec	14, 11	24, 26	~370 avg

- **Precinct South (excluding the Oct 2023 outlier):** Stationary. Stays between 22–28 throughout 2021–2023. Average always near 25. Predictable and stable.